METABOLISM

Module map

- Module 06: Metabolism
- Connected lab: lab-06_metabolism.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 06

Metabolism

1. Energy and thermodynamics
2. ATP as cellular coupling
3. Enzymes and activation energy
4. Regulation by conditions and inhibitors
5. Photosynthesis and respiration as linked pathways

Lab evidence

lab-06_metabolism.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

METABOLISM

Learning objectives

- Apply thermodynamic ideas to living cells.
- Explain ATP as an energy-coupling molecule.
- Describe how enzymes change reaction rates.
- Predict how environmental conditions affect enzyme activity.
- Connect photosynthesis and cellular respiration in matter and energy flow.

OBJECTIVE 1
Apply thermodynamic ideas to living cells.

OBJECTIVE 2
Explain ATP as an energy-coupling molecule.

OBJECTIVE 3
Describe how enzymes change reaction rates.

OBJECTIVE 4
Predict how environmental conditions affect enzyme activity.

OBJECTIVE 5
Connect photosynthesis and cellular respiration in matter and energy flow.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

METABOLISM

Topic sequence

- Energy and thermodynamics: Cells transform energy while obeying thermodynamic constraints.
- ATP as cellular coupling: ATP couples energy-releasing reactions to energy-requiring cellular work.
- Enzymes and activation energy: Enzymes lower activation energy without being consumed.
- Regulation by conditions and inhibitors: Temperature, pH, substrate concentration, and inhibitors alter enzyme function.
- Photosynthesis and respiration as linked pathways: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

01
Energy and thermodynamics
Cells transform energy while obeying thermodynamic constraints.

02
ATP as cellular coupling
ATP couples energy-releasing reactions to energy-requiring cellular work.

03
Enzymes and activation energy
Enzymes lower activation energy without being consumed.

04
Regulation by conditions and inhibitors
Temperature, pH, substrate concentration, and inhibitors alter enzyme function.

05
Photosynthesis and respiration as linked pathways
Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

METABOLISM

Module 06: Metabolism Evidence Map

- Module focus: Metabolism
- Central claim: Metabolism explains energy transformations with ATP coupling, enzymes, and linked pathways.
- Key nodes to connect: Energy and thermodynamics, ATP as cellular coupling, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-06_metabolism.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

METABOLISM

Module 06: Metabolism Reasoning Sequence

- Module focus: Metabolism
- Trace the model: Energy and thermodynamics -> ATP as cellular coupling -> Enzymes and activation energy -> Regulation by conditions and inhibitors
- Inputs become outputs: Energy and thermodynamics, ATP as cellular coupling -> Apply thermodynamic ideas to living cells., Explain ATP as an energy-coupling molecule.
- Feedback check: lab-06_metabolism.md evidence checks whether students can explain enzymes and activation energy.
- Lab connection: lab-06_metabolism.md gives

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

METABOLISM

Terms as evidence handles

- Metabolism: All chemical reactions in a cell or organism.
- ATP: Cellular energy-coupling molecule.
- Activation energy: Energy required to start a reaction.
- Enzyme: Protein or RNA catalyst that speeds a reaction.
- Substrate: Reactant that binds an enzyme active site.
- Denaturation: Loss of shape that disrupts function.

METABOLISM
All chemical reactions in a cell or organism.

ATP
Cellular energy-coupling molecule.

ACTIVATION ENERGY
Energy required to start a reaction.

ENZYME
Protein or RNA catalyst that speeds a reaction.

SUBSTRATE
Reactant that binds an enzyme active site.

DENATURATION
Loss of shape that disrupts function.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

METABOLISM

Lab connection

- Lab file: lab-06_metabolism.md
- Lab evidence should test or illustrate: Enzymes and activation energy
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Energy and thermodynamics

EVIDENCE
lab-06_metabolism.md

CLAIM
Apply thermodynamic ideas to living cells.

Teaching note: Make the lab connection explicit before students encounter the activity.

METABOLISM

Contrast check

- Do not stop at: Cells transform energy while obeying thermodynamic constraints.
- Stronger explanation: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Cells transform energy while obeying thermodynamic constraints.

DEEPER
Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

METABOLISM

Module 06: Metabolism Retrieval and Lab Check

- Module focus: Metabolism
- Retrieval prompt: How do the laws of thermodynamics constrain cells?
- Answer check: Apply thermodynamic ideas to living cells.
- Required terms: Metabolism, ATP, Activation energy
- Lab connection: lab-06_metabolism.md; lab-06_metabolism.md supplies evidence for reaction-rate shifts under temperature, pH, and substrate changes.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

METABOLISM

Practice quiz bridge

- 1. What is the main role of ATP in a cell?
- 2. How does an enzyme speed up a chemical reaction?
- 3. Raising temperature far above normal can stop an enzyme from working because:
- 4. How are photosynthesis and cellular respiration related?

QUESTION 1

Key: B

What is the main role of ATP in a cell?

QUESTION 2

Key: A

How does an enzyme speed up a chemical reaction?

QUESTION 3

Key: D

Raising temperature far above normal can stop an enzyme from working because:

QUESTION 4

Key: C

How are photosynthesis and cellular respiration related?

Teaching note: Use the quiz as formative feedback. Answer key: 1=B, 2=A, 3=D, 4=C

METABOLISM

Synthesis and exit ticket

- Exit claim: explain energy and thermodynamics using evidence from lab-06_metabolism.md.
- Must include: Metabolism, ATP, Activation energy.
- Revision prompt: what evidence would change your explanation?

CLAIM
Energy and thermodynamics

EVIDENCE
lab-06_metabolism.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.